Cells & fuel cells

Make a chemical cell

1 Simple cells

Two different metals in electrolyte, connected externally. Chemical reaction supplies electrical energy.

Voltage depends on electrodes and electrolyte. Keep electrolyte/concentrationand temperature fixed when comparing metals.

Use lead polarity and reference convention to interpret reactivity data.

2 Batteries & recharge

Series voltages add: 3 × 1.5 V = **4.5 V.** Battery = two or more cells.

Non-rechargeable, e.g. alkaline: stops when reactant is used up.

Rechargeable: external current reverses chemical reactions. Compare mass, capacity, lifetime cost and recharge time.

Follow fuel-cell redox

3 Continuous fuel supply

External hydrogen and oxygen/air feed the cell. $2H_2 + O_2 \rightarrow 2H_2O$. Hydrogen oxidised; oxygen reduced.

Electrons travel through external circuit/load; ions through electrolyte.

Continues while supplied with reactants; refill fuel rather than reverse reaction to recharge.

4 Acidic half-equations

Hydrogen: $2H_2 \rightarrow 4H^+ + 4e^-$.

Oxygen: $O_2 + 4H^+ + 4e^- \rightarrow 2H_2O$.

Add and cancel H^+/electrons. Balance atoms and charge. Fuel cell generates current; electrolysis requires a power supply.

Evaluate the choice

5 If electrolyte is alkaline

Hydrogen: $2H_2 + 4OH^- \rightarrow 4H_2O + 4e^-$.

Oxygen: $O_2 + 2H_2O + 4e^- \rightarrow 4OH^-$.

Sum gives same overall reaction. Use the electrolyte specified; do not mix acidic and alkaline half-equations.

6 Fuel cell vs rechargeable

Fuel: water during use, quick refuelling possible; hydrogen storage, flammability and infrastructure matter.

Battery: recharge from grid, no exhaust; mass, charge time, finite life and materials matter.

Check hydrogen production/electricitysources, efficiency and manufacture. Neither is automatically zero-carbon overall.